one in which every single mode has the lowest possible energy. That's a unique state, which we call the *vacuum*. When quantum field theorists talk about the vacuum, they don't mean a machine that lifts dust off your floors, or even a region of interplanetary space devoid of matter. What they mean is "the lowest-energy state of your quantum field theory."

You might think that the quantum vacuum would be empty and boring, but it's actually a wild place. An electron in an atom has a lowest-energy state it can be in, but if we think about it as a wave function of the position of the electron, that function can still have an interesting shape. Likewise, the vacuum state in field theory can still have interesting structure if we ask about individual parts of the field.

The next energy level has a bit more going on, since we make it out of the next-highest energies of each mode. That gives us a bit of freedom; there can be states that are mostly short-wavelength modes, or states that are mostly long-wavelength modes, or any mixture. What they have in common is each mode is in its "first excited state," with just a bit more energy than the minimum.

Putting that together, the wave function for the first excited state of a quantum field theory looks exactly like that of a single particle, expressed as a function of momentum rather than position. There will generally be contributions from different wavelengths, which we interpret as different momenta in the particle wave function. Most important, this kind of state behaves in a particle-like way when we observe it: if we measure a bit of energy in one location (interpreted as "I just saw a particle there"), it becomes overwhelmingly probable that you will observe the same amount of energy nearby if you look a moment later, even if the wave function was originally all spread out. What you end up seeing is a localized vibration propagating in the field, leaving a track in an experimental detector just like a particle is supposed to do. If it looks like a particle and quacks like a particle, it makes sense to call it a particle.

Can we have a quantum-field-theory wave function that combines some modes in their lowest-energy states and some others in their first excited states? Sure—that would be a superposition of a zero-particle state and a one-particle state, giving a state without a definite number of particles.

As you might be prepared to guess, the next-highest energy wave functions of a quantum field theory look like the wave function of two particles. The story goes on for quantum field states representing three